

PROJECT WORK

CE2204: Water Resources
Engineering

**Gandhi Sagar Hydro
Power Plant**



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History & Background of the Gandhi Sagar Dam

The Gandhi Sagar Dam is one of the four big dams constructed across the Chambal River and forms the focal point of the Chambal Valley Development Project—a collaborative effort between the Madhya Pradesh and Rajasthan state governments. This river development scheme on a large scale was one of the major projects initiated under India's First Five-Year Plan (1951–1956) soon after India achieved independence in 1947.

Situated in Madhya Pradesh's Mandsaur district, the dam is placed in a gorge reach of the Chambal River, which rises from the Vindhya Range at Mhow and moves north-northeast through Rajasthan and Madhya Pradesh to join the Yamuna River. The part of the river between 344 km and 440 km from its origin is marked by deep gorges—perfect for hydropower schemes because of the natural drop in elevation. Identifying the irrigation and hydroelectric potential of Chambal's 625-metre fall from Mhow to Kota, the government suggested a three-stage plan of development in 1953. The initial stage was to construct the Gandhi Sagar Dam as a storage and power-producing dam. Prime Minister Pandit Jawaharlal Nehru laid its foundation on 7 March 1954, and the main work, undertaken by Dwarka Das Agrawal & Associates, was done in 1960.

The dam, which is a masonry gravity structure 62.17 meters in height, accommodates a 115 MW hydroelectric power plant at its bottom, with five generating units each of 23 MW capacity. The released water after generation flows downstream to Kota Barrage, 104 km away in the state of Rajasthan, allowing irrigation for more than 427,000 hectares of agricultural fields—benefiting both partner states.

The Gandhi Sagar Dam formed the basis of the second and third phases of the Chambal project, i.e., the Rana Pratap Sagar Dam and Jawahar Sagar Dam, respectively, further increasing the area's hydroelectric and irrigation potential.

Power Generation at Gandhi Sagar Dam

1. Power Generation Requirements

To produce hydroelectric power effectively, the following is required:

a. Water Source & Reservoir

- Reservoir: Potential energy is stored in the form of water at elevation. Gandhi Sagar Reservoir has a gross storage capacity of 7.32 billion cubic metres and a live storage of 6.79 billion cubic metres at Full Reservoir Level (FRL).
- Water Head: Water height above turbines. Gandhi Sagar Dam is 62.17 metres (204 ft) high—appropriate for medium-head hydroelectric schemes.

b. Intake Structures and Penstocks

- Direct and control water from the reservoir to turbines through penstocks—large pressure pipes.

c. Turbines

- Gandhi Sagar employs Francis turbines (most likely, considering its head and flow conditions).
- Six turbines are present:
 - Five of 23 MW
 - One of 27 MW
- The turbines transfer mechanical energy from water to kinetic energy.

d. Generators

- When driven by turbines, they transfer mechanical energy to electrical energy through electromagnetic induction.

e. Control Systems and Switchyard

- In order to operate the operations, transmission, and safety.
- Power is transmitted through transformers, increased in voltage, and delivered to the grid.

2. How Power is Generated at Gandhi Sagar

1. Water from the reservoir is released through penstocks.
2. The energy of falling water makes the turbine blades rotate.
3. Rotating turbines drive generators, which generate alternating current (AC) electricity.
4. Electricity is supplied to the nearby district initially and distributed to other regions of Madhya Pradesh and Rajasthan.
5. Downstream water is discharged to the Kota Barrage for irrigation.

3. Technical Specifications

Feature	Details
Dam Type	Masonry Gravity
Height	62.17 m (204 ft)
Length	514 m
Powerhouse Location	Toe of the dam, right bank
Installed Capacity	142 MW total
Turbines	5 x 23 MW + 1 x 27 MW
Water Flow Through Turbines	311.15 m³/s
Generation Cost	₹2.3 billion total project cost
Power Supply Area	MP & Rajasthan

4. Power Generation Output

- The dam generates around 564 GWh (Gigawatt hours) annually, under typical conditions.
- However, studies show that in only 5 years out of 50, the reservoir reached full storage capacity—affecting generation efficiency.
- Over 50 years, total generation from the Chambal Valley projects (including Gandhi Sagar) declined by 25% from expected figures, primarily due to inconsistent rainfall and reservoir fill levels.

5. Managing Seasonal & Demand Fluctuations

a. Seasonal Impact

- Monsoon Season: Higher water availability boosts electricity generation.
- Dry Season: Reduced inflow lowers generation potential.
- Gandhi Sagar, like other reservoirs, operates on seasonal water storage planning, ensuring enough water is retained post-monsoon for lean seasons.

b. Demand Management

- Electricity demand increases in summer (due to ACs, fans, irrigation).
- Decreases in monsoon and winter when irrigation and cooling loads are lower.

c. How Fluctuations Are Handled

1. Load Forecasting: Power utilities forecast seasonal demand and adjust dam water release accordingly.
2. Grid Connectivity: Gandhi Sagar is connected to a larger regional grid, so any shortfall or surplus can be balanced across regions.
3. Hydro-Peaking: Turbines can be started/stopped quickly, so power generation is ramped up during peak demand hours (morning/evening).
4. Pumped Storage Project (Upcoming):
 - A 1,440 MW Pumped Storage Hydroelectricity Project is under construction (as of 2022).
 - It will pump water back up to a higher reservoir during low demand periods and release it during peak demand, helping balance supply and demand like a giant battery.

6. Environmental & Secondary Benefits

- Power generation is eco-friendly—zero emissions.
- Reservoir supports fisheries, bird habitat, and eco-tourism.
- Fish capture is regulated seasonally (June 16 to August 15) to protect biodiversity.

Reservoir Operation:

1. Besides its main applications in hydroelectricity generation and irrigation, the Gandhi

Sagar Reservoir also serves a variety of secondary but important purposes:

a. Ecological Habitat & Biodiversity

- It is visited by numerous migratory and non-migratory birds throughout the year.
- It has been identified by the International Bird Life Agency (IBA) as per the "A4iii" criteria since it regularly holds more than 20,000 waterbirds, making it a significant bird conservation area.
- The surrounding region of the reservoir has varied flora and fauna, thus supporting regional ecological equilibrium.

b. Fisheries & Livelihood

- It is a resourceful source of inland fisheries, supporting livelihood and sustenance of the local fishing community.
- The calming and nutrient-rich waters sustain different species of freshwater fish, which is an addition to the local economy.

c. Tourism & Recreation

- Its beauty, along with the avian population, makes it a nature tourism spot, bird watching, and eco-tourism destination.
- Boating and picnicking are common activities that lure locals and tourists, particularly the migratory bird season.

d. Flood Control & Regulation of Water

- Regulating the flow of the Chambal River, the dam contributes to checking floods during the rainy season and ensuring water availability during the dry season.

It provides a constant and regulated discharge of water for downstream consumption, particularly irrigation through the Kota Barrage.

2. How Water is Managed in the Reservoir

a. Inflows

Water enters the reservoir mainly from:

- Chambal River
- Its tributaries: Shipra, Kalisindh, Rupniya, etc.
- Seasonal rainfall (avg ~860 mm/year)

All inflows are measured by river gauges and remote sensors. This data helps forecast:

- How much water will be available.
 - When to store, use, or release water.
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b. Decision-making for Storage & Release

Water levels are managed based on:

- Full Reservoir Level (FRL): 400 m
- Live storage: 6.79 billion m³ (usable water)
- Dead storage: Water below outlets—can't be used

They monitor:

- Water inflows
- Storage levels
- Weather forecasts
- Demand forecasts (irrigation & power)

A Reservoir Operation Schedule guides when and how much water should be released.

3. How is Water Released?

a. For Power Generation

- Water flows through penstocks to the turbines.
- The total turbine flow capacity is 311.15 m³/s.
- Controlled by gates at the intake structure.

b. For Flood Control

- The dam has:

- 10 gated spillway spans (max discharge: 21,238 m³/s)
- 9 river sluices (non-functional)
- During heavy rainfall, spillway gates are opened to safely discharge excess water and prevent overtopping.

c. For Irrigation

- Water released after power generation flows downstream to the Kota Barrage, 104 km away.
 - From there, it's distributed through canals to irrigate 427,000+ hectares of farmland in MP and Rajasthan.
 - Water release is planned seasonally, aligned with cropping cycles (Kharif & Rabi).
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4. Optimization Strategies

a. Coordinated Reservoir Operation

- Gandhi Sagar is part of a 3-tier dam system:
 1. Gandhi Sagar Dam
 2. Rana Pratap Sagar Dam (48 km downstream)
 3. Jawahar Sagar Dam
- They work in coordination, optimizing water use for:
 - Energy at each stage
 - Irrigation efficiency
 - Flood control

b. Real-Time Monitoring

- Authorities use:
 - Telemetry data
 - Satellite rainfall predictions
 - River basin models
- Helps make dynamic decisions on:
 - Water conservation in dry spells

- Quick releases during floods

c. Pumped Storage System (Upcoming)

- A 1,440 MW pumped-storage project is underway.
 - During low demand, water is pumped back up to a higher reservoir.
 - During peak demand, it's released again for extra power.
 - This makes water and energy use highly efficient.
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References

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